

Technical note: a counterexample to the Rockafellar sum conjecture on c_0

Abstract

We give a counterexample to the Rockafellar sum conjecture on c_0 . There exists a maximally monotone operator $M : c_0 \rightrightarrows \ell^1$ such that, for the closed ball C_0 of radius 12 centered at the origin, $\text{dom } M \cap \text{int } C_0 \neq \emptyset$ but $M + N_{C_0}$ is not maximally monotone. We construct a monotone seed set using interval indicators and Lorentz-type coordinates, then control the local duality pairing of every point in its monotone polar. Zorn's lemma supplies a maximal monotone extension. The origin is monotonically related to the graph of the resulting sum but does not belong to that graph.

1 Introduction

Rockafellar's 1970 sum theorem [5] states that, on a real reflexive Banach space X , two maximally monotone operators $A, B : X \rightrightarrows X^*$ have a maximally monotone sum whenever

$$\text{dom } A \cap \text{int } \text{dom } B \neq \emptyset.$$

The Rockafellar sum conjecture asks whether the same implication holds without the reflexivity assumption. Throughout this note, interior is taken in the norm topology.

We give a counterexample on the nonreflexive Banach space c_0 . Specifically, Theorem 10 provides a maximally monotone operator $M : c_0 \rightrightarrows \ell^1$ and the closed ball $C_0 \subset c_0$ of radius 12 centered at the origin such that

$$\text{dom } M \cap \text{int } C_0 \neq \emptyset, \quad M + N_{C_0} \text{ is not maximally monotone.}$$

Thus the interior condition does not suffice in arbitrary Banach spaces, even when the second operator is the normal cone of a closed ball.

The Hilbert-space range theory underlying the resolvent approach goes back to Minty [3] in 1962. In 1970, Rockafellar [4] proved that subdifferentials of proper lower semicontinuous convex functions are maximally monotone on general Banach spaces; normal-cone operators are a special case. Bauschke, Borwein, Wang, and Yao [1] constructed maximally monotone operators that fail Gossez's dense-type (D) property in 2012. This failure alone does not constitute a counterexample to the sum conjecture.

Positive sum results beyond reflexive spaces also hold under additional structure. For example, Borwein and Yao [2] proved in 2014 that the above interior condition suffices when A is of Fitzpatrick–Phelps–Verona (FPV) type and $\text{dom } A$ is convex. Their paper described the unrestricted Banach-space sum problem as open at that time.

Our proof starts with an explicitly constructed monotone *seed set* $S \subseteq c_0 \times \ell^1$. Interval indicators provide Lorentz-type coordinates for constructing S and controlling its monotone polar S^μ . Once the three conditions of Lemma 1 are verified, Zorn's lemma supplies a maximal monotone extension G with $S \subseteq G \subseteq S^\mu$. The argument requires neither uniqueness nor a characterization of the resulting maximal extension.

2 From a Monotone Seed Set to a Sum Counterexample

We first isolate the properties of a monotone seed set that suffice for the counterexample. The reduction uses a maximal monotone extension of the seed and the normal cone of a closed ball.

2.1 Notation and preliminaries

Let X be a real Banach space with dual X^* , and set $Z = X \times X^*$. For all primal vectors in X , we write $\mathbf{x}, \mathbf{y} \in X$, and for all dual vectors in X^* , we write $\mathbf{x}^*, \mathbf{y}^* \in X^*$. A generic product-space point is written $\mathbf{z} = (\mathbf{x}, \mathbf{x}^*)$, while its two components are denoted by $\mathbf{z}_X = \mathbf{x}$ and $\mathbf{z}_{X^*} = \mathbf{x}^*$, respectively. For $\mathbf{z} = (\mathbf{x}, \mathbf{x}^*)$ and $\mathbf{w} = (\mathbf{y}, \mathbf{y}^*)$ in Z , let us define the *duality coupling* $c : Z \rightarrow \mathbb{R}$ and the symmetric bilinear form $B : Z \times Z \rightarrow \mathbb{R}$ by

$$c(\mathbf{z}) = \langle \mathbf{x}, \mathbf{x}^* \rangle \quad \text{and} \quad B(\mathbf{z}, \mathbf{w}) = \langle \mathbf{x}, \mathbf{y}^* \rangle + \langle \mathbf{y}, \mathbf{x}^* \rangle.$$

Then

$$c(\mathbf{z} - \mathbf{w}) = c(\mathbf{z}) + c(\mathbf{w}) - B(\mathbf{z}, \mathbf{w}).$$

With the duality coupling c , we know a set $S \subseteq Z$ is *monotone* if and only if $c(\mathbf{z} - \mathbf{w}) \geq 0$ for every pair of $\mathbf{z}, \mathbf{w} \in S$. Its *monotone polar* S^μ is then defined as:

$$S^\mu = \{\mathbf{w} \in Z : c(\mathbf{w} - \mathbf{z}) \geq 0 \text{ for every } \mathbf{z} \in S\}.$$

Thus S^μ consists of all points compatible with every point of S under the monotonicity inequality. A monotone set is maximally monotone if it has no proper monotone extension, or equivalently when $S = S^\mu$.

For a nonempty closed convex set $D \subseteq X$, its normal-cone operator is

$$N_D(\mathbf{x}) = \begin{cases} \{\boldsymbol{\nu}^* \in X^* : \langle \mathbf{y} - \mathbf{x}, \boldsymbol{\nu}^* \rangle \leq 0 \text{ for all } \mathbf{y} \in D\}, & \mathbf{x} \in D, \\ \emptyset, & \mathbf{x} \notin D. \end{cases}$$

The operator N_D is maximally monotone: it is the subdifferential of the proper lower semicontinuous convex indicator of D , so Rockafellar's subdifferential theorem applies [4].

2.2 The seed-to-counterexample reduction

The next lemma reduces the construction to three conditions on a monotone seed and its polar. We write $\|\cdot\|_X$ for the norm of X .

Lemma 1. *Suppose that $S \subseteq Z$ is a monotone set, and there exist $0 < R_0 < R$ such that:*

- (i) *There exists a point $\mathbf{z}_0 \in S$ with negative duality coupling, that is, $c(\mathbf{z}_0) < 0$.*
- (ii) *There exists $\mathbf{w}_0 = (\mathbf{y}, \mathbf{y}^*) \in S$ such that $\|\mathbf{y}\|_X < R_0$.*
- (iii) *For every $\mathbf{z} = (\mathbf{x}, \mathbf{x}^*) \in S^\mu$, if $\|\mathbf{x}\|_X < R$, then $c(\mathbf{z}) \geq 0$.*

Set $C_0 = \{\mathbf{x} \in X : \|\mathbf{x}\|_X \leq R_0\}$, then there is a maximally monotone operator $M : X \rightrightarrows X^$ such that $\text{dom } M \cap \text{int } C_0 \neq \emptyset$, but $M + N_{C_0}$ is not maximally monotone.*

Proof. Consider the partially ordered set ordered by inclusion:

$$\mathcal{F} = \{T \subseteq Z : S \subseteq T, T \text{ is monotone}\}.$$

It is nonempty because $S \in \mathcal{F}$. The union of any nonempty chain in $\mathcal{F}$ is again monotone and the union contains S , indicating that it is an upper bound in $\mathcal{F}$. For empty chains, they have

S as an upper bound. Therefore, by Zorn's lemma, there is a maximal element $G \in \mathcal{F}$. Any monotone proper extension of G would still contain S , so G is maximally monotone in the full space Z . Then define the operator induced by G as

$$M\mathbf{x} = \{\mathbf{x}^* \in X^* : (\mathbf{x}, \mathbf{x}^*) \in G\}.$$

Then $\text{gra } M = G$. Since G is monotone and contains S , we have $S \subseteq G \subseteq S^\mu$.

Denote $\mathbf{0}_Z = (\mathbf{0}, \mathbf{0}^*)$. Condition (i) implies $\mathbf{0}_Z \notin G$, since otherwise monotonicity with $\mathbf{z}_0$ would give $c(\mathbf{z}_0) = c(\mathbf{0}_Z - \mathbf{z}_0) \geq 0$, which is a contradiction. Condition (ii) and $\mathbf{w}_0 \in S \subseteq G$ give $\mathbf{y} \in \text{dom } M \cap \text{int } C_0$, and hence $\text{dom } M \cap \text{int } C_0 \neq \emptyset$.

Now, take any point of the sum graph, and choose a decomposition

$$(\mathbf{x}, \mathbf{x}^* + \boldsymbol{\nu}^*) \in \text{gra}(M + N_{C_0}), \quad \mathbf{x}^* \in M\mathbf{x}, \quad \boldsymbol{\nu}^* \in N_{C_0}(\mathbf{x}).$$

Since $\mathbf{x} \in C_0$, we have $\|\mathbf{x}\|_X \leq R_0 < R$. By $G \subseteq S^\mu$ and condition (iii), we have $\langle \mathbf{x}, \mathbf{x}^* \rangle \geq 0$. As $\mathbf{0} \in C_0$, the normal-cone inequality with $\mathbf{y} = \mathbf{0}$ gives $\langle -\mathbf{x}, \boldsymbol{\nu}^* \rangle \leq 0$ and hence

$$c(\mathbf{0}_Z - (\mathbf{x}, \mathbf{x}^* + \boldsymbol{\nu}^*)) = \langle \mathbf{x}, \mathbf{x}^* + \boldsymbol{\nu}^* \rangle \geq 0.$$

Thus $\mathbf{0}_Z$ is monotonically related to $\text{gra}(M + N_{C_0})$. However, if $\text{gra}(M + N_{C_0})$ is maximal, then $\mathbf{0} \in \text{int } C_0$ implies $N_{C_0}(\mathbf{0}) = \{\mathbf{0}^*\}$, so $\mathbf{0}_Z \in \text{gra}(M + N_{C_0})$ would force $\mathbf{0}_Z \in G$, which yields a contradiction as Condition (i) requires $\mathbf{0}_Z \notin G$. Therefore $\text{gra}(M + N_{C_0})$ cannot be maximally monotone, which completes the proof of this lemma. $\square$

The remaining task is to construct one seed S satisfying this lemma. There is no need to describe every point of the eventual maximal extension.

3 Lorentz geometry and the seed template

3.1 A Lorentz-type geometry framework

Lorentz-type coordinates turn the monotonicity condition into a Lipschitz bound. From now on, we work with

$$X = c_0, \quad X^* = \ell^1, \quad Z = c_0 \times \ell^1, \quad \langle \mathbf{x}, \mathbf{a}^* \rangle = \sum_{j \geq 1} x_j a_j^*,$$

where $c_0 := \{\mathbf{x} = (x_j)_{j \geq 1} : x_j \rightarrow 0\}$ with norm $\|\mathbf{x}\|_\infty := \sup_{j \geq 1} |x_j|$ is the nonreflexive Banach space of real null sequences, and $\ell^1 := \{\mathbf{a}^* = (a_j^*)_{j \geq 1} : \sum_{j \geq 1} |a_j^*| < \infty\}$ with norm $\|\mathbf{a}^*\|_1 := \sum_{j \geq 1} |a_j^*|$ is the nonreflexive Banach space of real absolutely summable sequences. Sequences are bold, their scalar entries are not, and a star distinguishes dual from primal sequences.

We first specify the geometry we need and leave its precise realization to Section 4. In particular, we seek a linear subspace $C \subseteq Z$ that contains the seed set S , a real Hilbert space $\mathcal{H}$ that will be used for introducing the negative duality coupling in Lemma 1, and a linear *injective* map

$$\Lambda : C \rightarrow \mathbb{R} \oplus \mathcal{H}, \quad \Lambda \mathbf{z} = (P(\mathbf{z}), N(\mathbf{z})), \quad (3.1)$$

with $P : C \rightarrow \mathbb{R}$ and $N : C \rightarrow \mathcal{H}$, such that the following identity holds for all $\mathbf{z} \in C$:

$$c(\mathbf{z}) = P(\mathbf{z})^2 - \|N(\mathbf{z})\|_{\mathcal{H}}^2 \quad (3.2)$$

Because C is linear and P, N are linear, applying this identity to $\mathbf{z} - \mathbf{w} \in C$ gives

$$c(\mathbf{z} - \mathbf{w}) = (P(\mathbf{z}) - P(\mathbf{w}))^2 - \|N(\mathbf{z}) - N(\mathbf{w})\|_{\mathcal{H}}^2, \quad \forall \mathbf{z}, \mathbf{w} \in C. \quad (3.3)$$

Thus no additional polarization is required for the difference identity. The difference-of-squares form (3.3) is what is meant here by *Lorentz coordinates*. It has three immediate consequences:

$$\begin{aligned} \|N(\mathbf{z}) - N(\mathbf{w})\|_{\mathcal{H}} \leq |P(\mathbf{z}) - P(\mathbf{w})| &\rightarrow c(\mathbf{z} - \mathbf{w}) \geq 0, \\ \|N(\mathbf{z})\|_{\mathcal{H}} \leq |P(\mathbf{z})| &\rightarrow c(\mathbf{z}) \geq 0, \\ P(\mathbf{z}) = 0, \quad N(\mathbf{z}) \neq \mathbf{0}_{\mathcal{H}} &\rightarrow c(\mathbf{z}) < 0. \end{aligned}$$

These observations will be used frequently in the later discussion. In the following sections, we will reserve the uppercase P, N for the maps. For the coordinates of a particular point $\mathbf{w} \in C$, we will use lowercase letters:

$$\Lambda \mathbf{w} = (p, \mathbf{n}), \quad p = P(\mathbf{w}) \in \mathbb{R}, \quad \mathbf{n} = N(\mathbf{w}) \in \mathcal{H}.$$

Finally, the injectivity of Λ means that these coordinates determine $\mathbf{w}$ uniquely when $(p, \mathbf{n}) \in \Lambda(C)$. It does not assert that every pair in $\mathbb{R} \oplus \mathcal{H}$ has a preimage.

3.2 The seed template

We specify the seed's form before constructing its components. We seek points $\mathbf{z}_0, \mathbf{v} \in C$ with $P(\mathbf{z}_0) = 0$ and $P(\mathbf{v}) = 1$, together with a sequence $(\mathbf{m}_i)_{i \geq 1} \subseteq C$. Then we will construct the seed set of the form

$$\gamma(p) = \mathbf{z}_0 + p\mathbf{v} \quad (p \leq \tfrac{1}{2}), \quad S = \{\gamma(p) : p \leq \tfrac{1}{2}\} \cup \{\mathbf{m}_i : i \geq 1\} \cup \{2\mathbf{v}\}, \quad (3.4)$$

where $\gamma(p), p \leq 1/2$ is a parameterized half-line in C . Roughly speaking, the point $\mathbf{z}_0 \in C$ will correspond to the negative-duality-coupling point $\mathbf{z}_0$ required by Lemma 1(i). The point $2\mathbf{v}$ will serve as the point $\mathbf{w}_0$ in Lemma 1(ii) that guarantees the constraint qualification. And finally, the sequence $\mathbf{m}_i$ is used to force the monotone polar S^μ to stay in C , which is key to verifying Lemma 1(iii).

4 Realization of the Lorentz framework

We now realize the framework of Section 3.1. The operator V supplies a Hilbert-space square norm, and the operator A places this quadratic form into the c_0 - ℓ^1 pairing.

4.1 Interval indicators and an injective Hilbert space map

Choose a *non-repeating enumeration* $(t_j)_{j \geq 1}$ of $\mathbb{Q} \cap (0, 1)$ with

$$t_1 = \frac{1}{2}, \quad t_2 = \frac{1}{4}.$$

For every pair of integers $i, k \geq 1$ and every $\varepsilon > 0$, there is an index $j > k$ such that

$$0 < |t_j - t_i| < \varepsilon. \quad (4.1)$$

Indeed, any neighborhood of t_i contains infinitely many rational numbers in $(0, 1)$, whereas the first k indices exclude only finitely many of them.

Let $H = L^2(0, 1)$ be the real Hilbert space of equivalence classes of square-integrable measurable functions on $(0, 1)$. Two functions are viewed as the same point in H if they differ only on a set of Lebesgue measure zero, and the zero element in H , denoted by 0_H , represents

the functions that equal 0 almost everywhere on $(0, 1)$. The inner product and norm on the Hilbert space H are given by

$$\langle f, g \rangle_H := \int_0^1 f(s)g(s) ds, \quad \|f\|_H := \left(\int_0^1 |f(s)|^2 ds \right)^{1/2}.$$

For each $j \geq 1$, define the interval indicator function

$$\phi_j := \mathbf{1}_{(0, t_j)} \in H.$$

Note that these are the 0–1 indicators so that $\phi_j(x) = 1$ only when $x \in (0, t_j)$, not the extended-real-valued indicators in convex analysis. Therefore, direct integration gives

$$\langle \phi_j, \phi_k \rangle_H = \min\{t_j, t_k\}, \quad \|\phi_j - \phi_k\|_H = \sqrt{|t_j - t_k|}. \quad (4.2)$$

For any $\mathbf{a}^* = (a_j^*)_{j \geq 1} \in \ell^1$, define the linear operator $V : \ell^1 \rightarrow H$ such that

$$V\mathbf{a}^* := \sum_{j \geq 1} a_j^* \phi_j \in H. \quad (4.3)$$

Since $\|\phi_j\|_H \leq 1$, this series converges absolutely in H , and hence $\|V\mathbf{a}^*\|_H \leq \|\mathbf{a}^*\|_1$.

Lemma 2 (Injectivity). *For $\mathbf{a}^* \in \ell^1$, $V\mathbf{a}^* = 0_H$ implies $\mathbf{a}^* = \mathbf{0}^*$.*

Proof. Suppose $V\mathbf{a}^* = 0_H$ and consider the representative

$$F(s) = \sum_{j \geq 1} a_j^* \mathbf{1}_{(0, t_j)}(s), \quad 0 < s < 1.$$

The series converges uniformly and absolutely because $\sum_j |a_j^*| < \infty$, hence

$$F = 0 \quad a.e.$$

Fix any $j_0 \geq 1$ and $\varepsilon > 0$. Choose a finite set $J \subseteq \mathbb{N}$ containing j_0 such that

$$\sum_{j \notin J} |a_j^*| < \varepsilon.$$

In particular, such a required set J exists because $\mathbf{a}^* \in \ell^1$ and is hence absolutely summable. Since $(t_j)_{j \geq 1}$ is non-repeating, we know $t_j \neq t_{j_0}$ whenever $j \neq j_0$, and we can choose $s_- < t_{j_0} < s_+$ sufficiently close to t_{j_0} such that $[s_-, s_+]$ contains no t_j with $j \in J \setminus \{j_0\}$. We can also require $F(s_-) = F(s_+) = 0$, because $F = 0$ a.e. It then follows that

$$0 = F(s_-) - F(s_+) = a_{j_0}^* + \sum_{\substack{j \notin J \\ s_- < t_j \leq s_+}} a_j^*.$$

Therefore, through standard triangle inequality, we have

$$|a_{j_0}^*| \leq \sum_{\substack{j \notin J \\ s_- < t_j \leq s_+}} |a_j^*| \leq \sum_{j \notin J} |a_j^*| < \varepsilon.$$

Since ε was arbitrary, we can drive it to zero and obtain $a_{j_0}^* = 0$. As the index j_0 is also arbitrarily selected, we have $\mathbf{a}^* = \mathbf{0}^*$. $\square$

4.2 Encoding the quadratic form in the duality pairing

Expanding the Hilbert norm in (4.3) gives

$$\|V\mathbf{a}^*\|_H^2 = \sum_{j \geq 1} t_j (a_j^*)^2 + 2 \sum_{j < k} \min\{t_j, t_k\} a_j^* a_k^*.$$

For any $\mathbf{a}^* \in \ell^1$, define the linear operator $A : \ell^1 \rightarrow c_0$ coordinatewise by

$$(A\mathbf{a}^*)_j := t_j a_j^* + 2 \sum_{k > j} \min\{t_j, t_k\} a_k^*, \quad j \geq 1. \quad (4.4)$$

This formula is obtained by grouping each cross term $2 \min\{t_j, t_k\} a_j^* a_k^*$ with its smaller index $j < k$ and then, together with the diagonal term $t_j (a_j^*)^2$, factoring out a_j^* .

Proposition 3. *Equation 4.4 defines a bounded linear operator $A : \ell^1 \rightarrow c_0$ with $\|A\| \leq 2$. For all pairs of $\mathbf{a}^*, \mathbf{b}^* \in \ell^1$, the following identities hold:*

$$\langle A\mathbf{a}^*, \mathbf{a}^* \rangle = \|V\mathbf{a}^*\|_H^2, \quad (4.5)$$

$$\langle A\mathbf{a}^*, \mathbf{b}^* \rangle + \langle A\mathbf{b}^*, \mathbf{a}^* \rangle = 2 \langle V\mathbf{a}^*, V\mathbf{b}^* \rangle_H. \quad (4.6)$$

Proof. First, it is straightforward to see that the operator A is linear. Then because $\mathbf{a}^* \in \ell^1$ and $t_i \leq 1, \forall i$, for every $j \geq 1$, we have

$$|(A\mathbf{a}^*)_j| \leq |a_j^*| + 2 \sum_{k > j} |a_k^*| \leq 2 \sum_{k \geq j} |a_k^*| \rightarrow 0,$$

which indicates that $A\mathbf{a}^* \in c_0$ and $\|A\mathbf{a}^*\|_\infty \leq 2\|\mathbf{a}^*\|_1$. Moreover, (4.5) holds because A is exactly constructed to let it hold. We only need to notice that all the rearrangements in constructing A are justified by $\sum_{j,k} |a_j^* a_k^*| = \|\mathbf{a}^*\|_1^2 < \infty$ and $\min\{t_j, t_k\} \leq 1$.

Finally, apply (4.5) to $\mathbf{a}^* + \mathbf{b}^*$ and subtract its instances for $\mathbf{a}^*$ and $\mathbf{b}^*$. Expanding the two sides leaves precisely (4.6). This is the standard real quadratic-form polarization trick. $\square$

4.3 The subspace C and its Lorentz coordinates

Now we are ready to present the Lorentz geometry framework needed for our construction. For each $j \geq 1$, write $\mathbf{e}_j \in c_0$ for the j th standard unit sequence and $\mathbf{e}_j^* \in \ell^1$ for the corresponding dual unit sequence. They have the same numerical entries but they lie in different spaces. For each pair of j, k , we have $\langle \mathbf{e}_j, \mathbf{e}_k^* \rangle = \delta_{jk}$, which takes value 1 when $j = k$ and 0 otherwise. For any $\mathbf{a}^* \in \ell^1$ and $\tau \in \mathbb{R}$, define

$$\Psi(\mathbf{a}^*, \tau) := (-A\mathbf{a}^* + \tau \mathbf{e}_1, \mathbf{a}^*) \in Z, \quad C := \text{ran } \Psi,$$

where $\text{ran } \Psi$ denotes the range of Ψ , that is,

$$\text{ran } \Psi = \left\{ \Psi(\mathbf{a}^*, \tau) : \mathbf{a}^* \in \ell^1, \tau \in \mathbb{R} \right\}.$$

Because the map Ψ is linear, C is a linear subspace in Z . In particular, the parameters are unique in the sense that the second component determines the sequence $\mathbf{a}^*$, after which the first determines the scalar τ . Then let $\mathcal{H} := H \times \mathbb{R}$, and for $f \in H$, $r \in \mathbb{R}$, define

$$\|(f, r)\|_{\mathcal{H}}^2 := \|f\|_H^2 + r^2$$

For any $\mathbf{z} = \Psi(\mathbf{a}^*, \tau) \in C$, define

$$P(\mathbf{z}) := \frac{\tau + a_1^*}{2}, \quad N(\mathbf{z}) := \left(V\mathbf{a}^*, \frac{\tau - a_1^*}{2} \right) \in \mathcal{H}, \quad (4.7)$$

These maps provide the Lorentz coordinates of Section 3.1. The unique parameter representation makes the definitions unambiguous. More explicitly, if

$$\Lambda \mathbf{z} = (P(\mathbf{z}), N(\mathbf{z})) = (p, \mathbf{n})$$

for some $\mathbf{z} \in C$, see (3.1), then the H -component of $\mathbf{n}$ is $V\mathbf{a}^*$. Since V is injective, this uniquely determines $\mathbf{a}^*$, and then

$$\tau = 2p - a_1^*$$

can be uniquely determined as well.

Proposition 4. *The maps P, N are linear, and $\Lambda \mathbf{z} = (P(\mathbf{z}), N(\mathbf{z}))$ is injective on C . The identities (3.2) and (3.3) hold on C .*

Proof. For $\mathbf{z} = \Psi(\mathbf{a}^*, \tau) \in C$, the pairing identity (4.5) gives

$$\begin{aligned} c(\mathbf{z}) &= \langle -A\mathbf{a}^* + \tau \mathbf{e}_1, \mathbf{a}^* \rangle = -\|V\mathbf{a}^*\|_H^2 + \tau a_1^* \\ &= \left(\frac{\tau + a_1^*}{2} \right)^2 - \left[\|V\mathbf{a}^*\|_H^2 + \left(\frac{\tau - a_1^*}{2} \right)^2 \right] = P(\mathbf{z})^2 - \|N(\mathbf{z})\|_{\mathcal{H}}^2, \end{aligned}$$

which proves (3.2). Then applying the energy identity to $\mathbf{z} - \mathbf{w} \in C$ and using linearity proves the difference identity (3.3). Regarding injectivity, if $\Lambda \mathbf{z} = 0$, then $V\mathbf{a}^* = 0$. Lemma 2 gives $\mathbf{a}^* = \mathbf{0}^*$, and $P(\mathbf{z}) = 0$ then gives $\tau = 0$. Thus $\mathbf{z} = \mathbf{0}_Z$, which completes the proof. $\square$

The construction also makes membership in C explicit, as presented below.

Lemma 5 (Carrier membership). *For every $\mathbf{w} = (\mathbf{x}, \mathbf{x}^*) \in c_0 \times \ell^1$, we have*

$$\mathbf{w} \in C \iff \mathbf{x} + A\mathbf{x}^* \in \mathbb{R}\mathbf{e}_1,$$

or equivalently, $(\mathbf{x} + A\mathbf{x}^)_i = 0$ for every $i \geq 2$.*

Proof. The equality $\mathbf{w} = \Psi(\mathbf{a}^*, \tau)$ first forces $\mathbf{a}^* = \mathbf{x}^*$ and hence $\mathbf{x} = -A\mathbf{x}^* + \tau \mathbf{e}_1$. $\square$

5 Realization and verification of the seed set

We construct $\mathbf{z}_0$, $\mathbf{v}$, and $(\mathbf{m}_i)_{i \geq 1}$, then verify monotonicity and the three conditions of Lemma 1.

Construction of $\mathbf{v}$ and $\mathbf{z}_0$. By (4.1), we can choose an index $i_0 > 2$ such that

$$0 < |t_{i_0} - t_1| < \frac{1}{64^2}, \quad \mathbf{v} := \Psi(\mathbf{e}_1^* - \mathbf{e}_{i_0}^*, 1) \in C. \quad (5.1)$$

And we can choose an index $j_0 > 2$ satisfying

$$|t_{j_0} - t_2| < \frac{1}{16}, \quad 512\sqrt{|t_{j_0} - t_2|} < \frac{1}{64}, \quad \mathbf{z}_0 := 512\Psi(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*, 0) \in C.$$

Note that $i_0, j_0 > 2$ and non-repetition ensure $t_{i_0} \neq t_1, t_{j_0} \neq t_2$. Hence $\mathbf{v}, \mathbf{z}_0 \neq \mathbf{0}_Z$.

A detector needed for the construction of $(\mathbf{m}_i)_{i \geq 1}$. When $\mathbf{w} \notin C$, Lemma 5 provides a nonzero coordinate $(\mathbf{x} + A\mathbf{x}^*)_i$ for some $i \geq 2$. Since membership in a monotone polar is tested through $c(\mathbf{z} - \mathbf{w}) = c(\mathbf{z}) + c(\mathbf{w}) - B(\mathbf{z}, \mathbf{w})$, and our goal is to force $S^\mu \subseteq C$, we need to convert this coordinate information into a signal detectable through the bilinear form B , which will later be encoded into the sequence $(\mathbf{m}_j)_{j \geq 1}$. The following lemma performs this conversion while keeping the Lorentz displacement small.

Lemma 6 (Single-coordinate detector). *For integers $j > i \geq 2$, define*

$$\mathbf{h}(i, j) := \Psi(\mathbf{e}_i^* - \mathbf{e}_j^*, 0) = (-A(\mathbf{e}_i^* - \mathbf{e}_j^*), \mathbf{e}_i^* - \mathbf{e}_j^*) \in C.$$

Then we have

$$P(\mathbf{h}(i, j)) = 0, \quad N(\mathbf{h}(i, j)) = (\phi_i - \phi_j, 0). \quad (5.2)$$

For any $\mathbf{w} = (\mathbf{x}, \mathbf{x}^) \in Z$, write $\mathbf{r} = \mathbf{x} + A\mathbf{x}^* \in c_0$, then it holds that*

$$B(\mathbf{w}, \mathbf{h}(i, j)) = r_i - r_j - 2\langle V\mathbf{x}^*, \phi_i - \phi_j \rangle_H.$$

Consequently, for fixed i , along any sequence of indices $j \rightarrow \infty$ such that $t_j \rightarrow t_i$, we have

$$\|N(\mathbf{h}(i, j))\|_{\mathcal{H}} \rightarrow 0, \quad B(\mathbf{w}, \mathbf{h}(i, j)) \rightarrow r_i = (\mathbf{x} + A\mathbf{x}^*)_i. \quad (5.3)$$

Proof. Denote $\mathbf{a}^* = \mathbf{e}_i^* - \mathbf{e}_j^* \in \ell^1$. Since $i, j \geq 2$, we have $a_1^* = 0$. Then (4.7) and $V\mathbf{e}_k^* = \phi_k$ give (5.2). By the definition of the bilinear form B , we have

$$B(\mathbf{w}, \mathbf{h}(i, j)) = \langle \mathbf{x}, \mathbf{a}^* \rangle - \langle A\mathbf{a}^*, \mathbf{x}^* \rangle.$$

Apply (4.6) to $\mathbf{a}^*, \mathbf{x}^* \in \ell^1$ and rearrange:

$$-\langle A\mathbf{a}^*, \mathbf{x}^* \rangle = \langle A\mathbf{x}^*, \mathbf{a}^* \rangle - 2\langle V\mathbf{x}^*, V\mathbf{a}^* \rangle_H.$$

Therefore

$$\begin{aligned} B(\mathbf{w}, \mathbf{h}(i, j)) &= \langle \mathbf{x} + A\mathbf{x}^*, \mathbf{a}^* \rangle - 2\langle V\mathbf{x}^*, V\mathbf{a}^* \rangle_H \\ &= \langle \mathbf{r}, \mathbf{e}_i^* - \mathbf{e}_j^* \rangle - 2\langle V\mathbf{x}^*, \phi_i - \phi_j \rangle_H \\ &= r_i - r_j - 2\langle V\mathbf{x}^*, \phi_i - \phi_j \rangle_H. \end{aligned}$$

Now $r_j \rightarrow 0$ because $\mathbf{r} \in c_0$. Also, by Cauchy–Schwarz and (4.2),

$$|\langle V\mathbf{x}^*, \phi_i - \phi_j \rangle_H| \leq \|V\mathbf{x}^*\|_H \sqrt{|t_i - t_j|} \rightarrow 0.$$

The same equality (4.2) gives the first limit in (5.3). Hence we complete the proof. $\square$

The construction of sequence $(\mathbf{m}_i)_{i \geq 1}$. Choose an arbitrary sequence

$$(q_i, \sigma_i)_{i \geq 1} \subseteq \{2, 3, \dots\} \times \{-1, 1\}$$

such that every pair of $(q, \sigma), q \geq 2, \sigma \in \{-1, +1\}$ appears infinitely many times. Such a sequence can be obtained, for example, by concatenating for $k = 1, 2, \dots$ the finite lists containing both $\{-1, +1\}$ for each $q = 2, \dots, k+1$. Set

$$s_i := 1 + 2^{-i}, \quad \rho_i := 2^{-(i+6)}, \quad i \geq 1.$$

Then fix each $i \geq 1$, we can use (4.1) and (4.2) to choose $k_i > r_i > \max\{i, q_i\}$ such that

$$s_i \|\phi_1 - \phi_{r_i}\|_H < \frac{\rho_i}{2}, \quad i \|\phi_{q_i} - \phi_{k_i}\|_H < \frac{\rho_i}{2}, \quad (5.4)$$

and then define the following points of C :

$$\xi_i := \Psi(s_i(\mathbf{e}_1^* - \mathbf{e}_{r_i}^*), s_i), \quad (5.5)$$

$$\mathbf{h}_i := \mathbf{h}(q_i, k_i), \quad \mathbf{m}_i := \xi_i + i\sigma_i \mathbf{h}_i. \quad (5.6)$$

These choices specify all components of the seed template (3.4).

The construction of S . With the above choices of $\mathbf{z}_0, \mathbf{v}$ and $(\mathbf{m}_i)_{i \geq 1}$, define

$$\gamma(p) := \mathbf{z}_0 + p\mathbf{v} \quad (p \leq \tfrac{1}{2}), \quad S := \{\gamma(p) : p \leq \tfrac{1}{2}\} \cup \{\mathbf{m}_i : i \geq 1\} \cup \{2\mathbf{v}\} \subseteq C. \quad (5.7)$$

We now verify monotonicity and the three conditions of Lemma 1, in that order.

5.1 The monotonicity of S

We verify the pairwise inequalities required for monotonicity.

Lemma 7. *The set S defined in (5.7) is monotone.*

Proof. By the Lorentz identity (3.3), it is enough to show $\|N(\mathbf{z}) - N(\mathbf{w})\|_{\mathcal{H}} \leq |P(\mathbf{z}) - P(\mathbf{w})|$ for all $\mathbf{z}, \mathbf{w} \in S$. From the definitions of $\mathbf{v}$ and $\mathbf{z}_0$, (4.7), and (4.2), we have

$$\begin{aligned} P(\mathbf{v}) &= 1, \quad N(\mathbf{v}) = (\phi_1 - \phi_{i_0}, 0), \quad 0 < \|N(\mathbf{v})\|_{\mathcal{H}} < \frac{1}{64}, \\ P(\mathbf{z}_0) &= 0, \quad N(\mathbf{z}_0) = (512(\phi_2 - \phi_{j_0}), 0), \quad 0 < \|N(\mathbf{z}_0)\|_{\mathcal{H}} < \frac{1}{64}. \end{aligned} \quad (5.8)$$

In particular, the two norms are strictly positive because the nonrepetition of the enumeration gives $t_{i_0} \neq t_1$ and $t_{j_0} \neq t_2$. For every $i \geq 1$, the parameter τ and the first dual coordinate of $\mathbf{m}_i$ are both equal to s_i , since $q_i, k_i \geq 2$. Hence

$$P(\mathbf{m}_i) = s_i, \quad N(\mathbf{m}_i) = (s_i(\phi_1 - \phi_{r_i}) + i\sigma_i(\phi_{q_i} - \phi_{k_i}), 0). \quad (5.9)$$

By the triangle inequality and (5.4), we have

$$\|N(\mathbf{m}_i)\|_{\mathcal{H}} < \rho_i < s_i - 1. \quad (5.10)$$

In particular, as $s_i \rightarrow 1$, $N(\mathbf{m}_i) \rightarrow \mathbf{0}_{\mathcal{H}}$ in $\mathcal{H}$. For $p \leq \frac{1}{2}$, the linearity of maps P and N gives

$$P(\gamma(p)) = p, \quad N(\gamma(p)) = N(\mathbf{z}_0) + pN(\mathbf{v}).$$

And we shall be repeatedly using the following estimate

$$\|N(\gamma(p))\|_{\mathcal{H}} \leq 1 - p, \quad p \leq \tfrac{1}{2}. \quad (5.11)$$

To see this, when $p \leq 0$, we have

$$\|N(\gamma(p))\|_{\mathcal{H}} \leq \|N(\mathbf{z}_0)\|_{\mathcal{H}} + (-p)\|N(\mathbf{v})\|_{\mathcal{H}} < \frac{1-p}{64} \leq 1-p.$$

When $0 \leq p \leq \frac{1}{2}$, it holds that

$$\|N(\gamma(p))\|_{\mathcal{H}} \leq \|N(\mathbf{z}_0)\|_{\mathcal{H}} + p\|N(\mathbf{v})\|_{\mathcal{H}} < \frac{3}{128} < \frac{1}{2} \leq 1-p.$$

Finally, $P(2\mathbf{v}) = 2$ and $\|N(2\mathbf{v})\|_{\mathcal{H}} = 2\|N(\mathbf{v})\|_{\mathcal{H}} < 1$. Identical points give equality in the monotonicity inequality. For distinct points, there are five cases.

Case 1: two points on the half-line. For distinct $p, p' \leq \frac{1}{2}$, we have

$$\|N(\gamma(p)) - N(\gamma(p'))\|_{\mathcal{H}} = \|N(\mathbf{v})\|_{\mathcal{H}} \cdot |p - p'| < |p - p'| = |P(\gamma(p)) - P(\gamma(p'))|.$$

Case 2: a half-line point and a point $\mathbf{m}_i$. For $p \leq \frac{1}{2}$, using (5.10) and (5.11) yields

$$\begin{aligned} \|N(\mathbf{m}_i) - N(\gamma(p))\|_{\mathcal{H}} &\leq \|N(\mathbf{m}_i)\|_{\mathcal{H}} + \|N(\gamma(p))\|_{\mathcal{H}} \\ &< (s_i - 1) + (1 - p) = s_i - p = |P(\mathbf{m}_i) - P(\gamma(p))|. \end{aligned}$$

Case 3: a half-line point and $2\mathbf{v}$. For $p \leq \frac{1}{2}$, the triangle inequality gives

$$\begin{aligned}\|N(2\mathbf{v}) - N(\gamma(p))\|_{\mathcal{H}} &\leq \|N(2\mathbf{v})\|_{\mathcal{H}} + \|N(\gamma(p))\|_{\mathcal{H}} \\ &< 1 + (1 - p) = 2 - p = |P(2\mathbf{v}) - P(\gamma(p))|.\end{aligned}$$

Case 4: two distinct $\mathbf{m}_i$ and $\mathbf{m}_j$. Suppose $i < j$ without loss of generality, by (5.10), we have

$$\begin{aligned}\|N(\mathbf{m}_i) - N(\mathbf{m}_j)\|_{\mathcal{H}} &\leq \|N(\mathbf{m}_i)\|_{\mathcal{H}} + \|N(\mathbf{m}_j)\|_{\mathcal{H}} \\ &< \rho_i + \rho_j \leq 2^{-(i+6)} + 2^{-(i+7)} < 2^{-(i+1)} \\ &\leq s_i - s_j = |P(\mathbf{m}_i) - P(\mathbf{m}_j)|.\end{aligned}$$

Case 5: a point $\mathbf{m}_i$ and $2\mathbf{v}$. Using (5.10), we have

$$\begin{aligned}\|N(2\mathbf{v}) - N(\mathbf{m}_i)\|_{\mathcal{H}} &\leq \|N(2\mathbf{v})\|_{\mathcal{H}} + \|N(\mathbf{m}_i)\|_{\mathcal{H}} \\ &< \frac{1}{2} \leq 2 - s_i = |P(2\mathbf{v}) - P(\mathbf{m}_i)|.\end{aligned}$$

Thus every pair of points in S satisfies the monotonicity inequality. $\square$

5.2 Condition (i): a negative duality pairing point

The point $\mathbf{z}_0 = \gamma(0)$ belongs to S . By (3.2) and (5.8), it holds that

$$c(\mathbf{z}_0) = P(\mathbf{z}_0)^2 - \|N(\mathbf{z}_0)\|_{\mathcal{H}}^2 = -\|N(\mathbf{z}_0)\|_{\mathcal{H}}^2 < 0. \quad (5.12)$$

Thus Lemma 1(i) holds at the currently constructed $\mathbf{z}_0$.

5.3 Condition (ii): a point in $\text{dom } M \cap \text{int } C_0$

The verified point is $\mathbf{w}_0 = 2\mathbf{v} \in S$. To locate its first component, use (5.1) and $\|A\| \leq 2$:

$$\begin{aligned}\|\mathbf{v}_X\|_{\infty} &= \|-A(\mathbf{e}_1^* - \mathbf{e}_{i_0}^*) + \mathbf{e}_1\|_{\infty} \\ &\leq \|A\| \|\mathbf{e}_1^* - \mathbf{e}_{i_0}^*\|_1 + \|\mathbf{e}_1\|_{\infty} \leq 2 \cdot 2 + 1 = 5.\end{aligned} \quad (5.13)$$

Therefore

$$\|(2\mathbf{v})_X\|_{\infty} \leq 10 < 12. \quad (5.14)$$

This verifies Lemma 1(ii) with $R_0 = 12$.

5.4 Condition (iii): local nonnegative duality pairing in S^{μ}

This condition concerns every point of S^{μ} , not only points of S . We first show that the polar lies in C .

Lemma 8. *The monotone polar satisfies $S^{\mu} \subseteq C$.*

Proof. Fix any $\mathbf{w} = (\mathbf{x}, \mathbf{x}^*) \in Z \setminus C$. Lemma 5 gives $q \geq 2$ such that $\alpha := (\mathbf{x} + A\mathbf{x}^*)_q \neq 0$. Choose $\sigma \in \{-1, 1\}$ so that $\sigma\alpha = |\alpha|$, and let $I \subseteq \mathbb{N}$ be the infinite set of indices for which $(q_i, \sigma_i) = (q, \sigma)$. In particular, such a required set I exists because $(q_i, \sigma_i)_{i \geq 1}$ is constructed so that every pair appears infinitely many times in the sequence. Then along $i \in I$, we have $k_i > i$, and (5.4) gives $\|\phi_q - \phi_{k_i}\|_H < \rho_i/(2i) \rightarrow 0$. By (4.2), $t_{k_i} \rightarrow t_q$. Hence Lemma 6 gives

$$B(\mathbf{w}, \mathbf{h}_i) \rightarrow \alpha \quad \text{as } i \rightarrow \infty, i \in I.$$

Set $\eta := |\alpha|/2 > 0$. For all sufficiently large $i \in I$, we have $\sigma_i B(\mathbf{w}, \mathbf{h}_i) \geq \eta$. Next, equip Z with the norm $\|\mathbf{z}\|_Z := \|\mathbf{z}_X\|_\infty + \|\mathbf{z}_{X^*}\|_1$. Then for $\mathbf{w} = (\mathbf{x}, \mathbf{x}^*)$ and $\mathbf{z} = (\mathbf{y}, \mathbf{y}^*)$ in Z , we have

$$\begin{aligned} |B(\mathbf{w}, \mathbf{z})| &\leq \|\mathbf{x}\|_\infty \|\mathbf{y}^*\|_1 + \|\mathbf{y}\|_\infty \|\mathbf{x}^*\|_1 \\ &\leq (\|\mathbf{x}\|_\infty + \|\mathbf{x}^*\|_1)(\|\mathbf{y}\|_\infty + \|\mathbf{y}^*\|_1) = \|\mathbf{w}\|_Z \|\mathbf{z}\|_Z. \end{aligned}$$

For $1 < s_i \leq 3/2$, $\|s_i(\mathbf{e}_1^* - \mathbf{e}_{r_i}^*)\|_1 = 2s_i \leq 3$. Expanding (5.5) with $\|A\| \leq 2$, we obtain

$$\begin{aligned} \|(\xi_i)_X\|_\infty &\leq 2 \cdot 3 + \frac{3}{2} = \frac{15}{2}, \\ \|(\xi_i)_{X^*}\|_1 &\leq 3, \quad \|\xi_i\|_Z \leq \frac{21}{2} < 11. \end{aligned}$$

Also, (5.9) and (5.10) yield

$$0 < c(\mathbf{m}_i) = s_i^2 - \|N(\mathbf{m}_i)\|_{\mathcal{H}}^2 \leq s_i^2 < 3.$$

Thus the duality pairing of $\mathbf{m}_i$ is uniformly bounded regardless of i . Using (5.6) and the linearity of B in its second argument, for all sufficiently large $i \in I$ we have

$$\begin{aligned} B(\mathbf{w}, \mathbf{m}_i) - c(\mathbf{m}_i) &= B(\mathbf{w}, \xi_i) + \sigma_i B(\mathbf{w}, \mathbf{h}_i) - c(\mathbf{m}_i) \\ &\geq -11\|\mathbf{w}\|_Z + \eta i - 3 \rightarrow +\infty, \quad \text{as } i \rightarrow \infty, i \in I \end{aligned} \tag{5.15}$$

Then if $\mathbf{w} \in S^\mu$, every $\mathbf{m}_i \in S$ would satisfy

$$0 \leq c(\mathbf{w} - \mathbf{m}_i) = c(\mathbf{w}) + c(\mathbf{m}_i) - B(\mathbf{w}, \mathbf{m}_i),$$

and hence $B(\mathbf{w}, \mathbf{m}_i) - c(\mathbf{m}_i) \leq c(\mathbf{w})$ for every i . This immediately contradicts (5.15), since $\mathbf{w}$ is fixed and $c(\mathbf{w})$ is finite. Thus $\mathbf{w} \notin S^\mu$ and we finish the proof of this lemma. $\square$

Proposition 9. *For every $\mathbf{w} = (\mathbf{x}, \mathbf{x}^*) \in S^\mu$ satisfying $\|\mathbf{x}\|_\infty < 16$, we have $c(\mathbf{w}) \geq 0$.*

Proof. By Lemma 8, we know $\mathbf{w} \in C$ and its Lorentz coordinates are well defined. Write

$$\Lambda \mathbf{w} = (p, \mathbf{n}) \quad \text{with } p = P(\mathbf{w}), \quad \mathbf{n} = N(\mathbf{w}).$$

We distinguish two ranges of p .

Case 1: $p > 1/2$. Compare $\mathbf{w}$ with each $\mathbf{m}_i \in S$. Because $\mathbf{w} \in S^\mu$, we know

$$0 \leq c(\mathbf{w} - \mathbf{m}_i) = (p - s_i)^2 - \|\mathbf{n} - N(\mathbf{m}_i)\|_{\mathcal{H}}^2.$$

Hence $\|\mathbf{n} - N(\mathbf{m}_i)\|_{\mathcal{H}} \leq |p - s_i|$. Together with the fact that $s_i \rightarrow 1$ and $N(\mathbf{m}_i) \rightarrow \mathbf{0}_{\mathcal{H}}$ in $\mathcal{H}$, this gives $\|\mathbf{n}\|_{\mathcal{H}} \leq |p - 1|$ and therefore

$$c(\mathbf{w}) = p^2 - \|\mathbf{n}\|_{\mathcal{H}}^2 \geq p^2 - (p - 1)^2 = 2p - 1 > 0.$$

Thus any $\mathbf{w}$ with negative duality pairing $c(\mathbf{w}) < 0$ must have $p \leq 1/2$.

Case 2: $p \leq 1/2$. Compare $\mathbf{w}$ with $\gamma(p) \in S$ that shares the same P -coordinate as $\mathbf{w}$ because $P(\gamma(p)) = p$, then $\mathbf{w} \in S^\mu$ gives

$$0 \leq c(\mathbf{w} - \gamma(p)) = -\|\mathbf{n} - N(\gamma(p))\|_{\mathcal{H}}^2 \leq 0.$$

Thus $\mathbf{n} = N(\gamma(p))$ and the injectivity of Λ gives $\mathbf{w} = \gamma(p) \in S$. In this case, $c(\gamma(p)) < 0$ gives

$$|p| < \|N(\mathbf{z}_0) + pN(\mathbf{v})\|_{\mathcal{H}} \leq \|N(\mathbf{z}_0)\|_{\mathcal{H}} + |p| \cdot \|N(\mathbf{v})\|_{\mathcal{H}}.$$

Since $\|N(\mathbf{z}_0)\|_{\mathcal{H}}, \|N(\mathbf{v})\|_{\mathcal{H}} < 1/64$, it follows that

$$|p| < \frac{\|N(\mathbf{z}_0)\|_{\mathcal{H}}}{1 - \|N(\mathbf{v})\|_{\mathcal{H}}} < \frac{1}{63}. \quad (5.16)$$

We now locate the candidate points $\gamma(p)$, $|p| \leq 1/63$ in the original c_0 variable, which equals

$$\gamma(p)_X = (\mathbf{z}_0)_X + p\mathbf{v}_X = -512A(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*) + p\mathbf{v}_X.$$

By (5.13), $\|\mathbf{v}_X\|_{\infty} \leq 5$. It remains to bound $\|(\mathbf{z}_0)_X\|_{\infty}$ from below. By definition (4.4) and our choice that $t_2 = 1/4$, we have

$$(A(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*))_2 = \frac{1}{4} - 2 \min\left\{\frac{1}{4}, t_{j_0}\right\}.$$

The choice $|t_{j_0} - 1/4| < 1/16$ implies $\min\{1/4, t_{j_0}\} > 3/16$. Hence $(A(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*))_2 < -1/8$, and

$$\|(\mathbf{z}_0)_X\|_{\infty} = 512\|A(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*)\|_{\infty} \geq 512|(A(\mathbf{e}_2^* - \mathbf{e}_{j_0}^*))_2| \geq 64.$$

Combining this with (5.13) and (5.16) gives

$$\|\gamma(p)_X\|_{\infty} \geq \|(\mathbf{z}_0)_X\|_{\infty} - |p|\|\mathbf{v}_X\|_{\infty} > 63, \quad \forall |p| < 1/63.$$

Combining the results of the two cases proves the proposition. $\square$

Taking $R = 16$, Proposition 9 verifies Lemma 1(iii).

5.5 Failure of the Rockafellar sum conjecture on c_0

The seed reduction now gives the claimed counterexample.

Theorem 10. *There exists a maximally monotone operator $M : c_0 \rightrightarrows \ell^1$ such that, for the closed ball in c_0*

$$C_0 = \{\mathbf{x} \in c_0 : \|\mathbf{x}\|_{\infty} \leq 12\},$$

its normal cone operator N_{C_0} is maximally monotone and

$$\text{dom } M \cap \text{int dom } N_{C_0} \neq \emptyset, \quad M + N_{C_0} \text{ is not maximally monotone.}$$

Proof. Apply Lemma 1 with $R_0 = 12$, $R = 16$, and $\mathbf{w}_0 = 2\mathbf{v}$. Take $\mathbf{z}_0$ in Lemma 1 the same as that in the seed monotone set construction. The preceding verifications establish all its hypotheses. It supplies a maximal monotone extension $G \supseteq S$ and an operator M with $\text{gra } M = G$. Moreover, $\text{dom } N_{C_0} = C_0$, since the normal cone is empty outside C_0 and contains $\mathbf{0}^*$ at every point of C_0 . Thus the interior condition has the stated form. The origin $\mathbf{0}_Z = (\mathbf{0}, \mathbf{0}^*)$ is the explicit witness to the non-maximality:

$$\mathbf{0}_Z \in (\text{gra}(M + N_{C_0}))^{\mu} \setminus \text{gra}(M + N_{C_0}),$$

which concludes the proof. $\square$

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